

Document Title	Criteria for the Intra-Hospital Transporting Team and Preparation of Pediatric Patients Transported from PICU and HD Areas
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Document Author	Kholoud Al Mukhaini, Khoula Al Ruqaishi, Nasra Al Ramadhani
Designation	
Document Reviewer	Abdul Rahman Ragab, PICU Team
Designation	
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Validated by		Approved by	
Name	Quality and Patient Safety	Name	Dr.Khoula Al Said
Designation		Designation	Director - Department of Child Health
Signature		Signature	 
Date		Date	25.032025

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Acronyms

PICU	Pediatric Intensive Care Unit
PHD	Pediatric High Dependency Unit
BLS	Basic Life Support
PALS	Pediatric Advanced Life Support

Document History and Version Control

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1	Initial Release	March 2025

Introduction

Transporting sick pediatric patients is not uncommon. The transport could be within the same health facility (intra-hospital) or between health facilities (inter-facility transport). Intra-hospital transport is a vital aspect of pediatric care, as it often involves moving patients between departments for procedures, diagnostics, or specialized interventions that cannot be performed in the PICU. Without a well-organized transport team, these transfers can pose significant risks, including deterioration of the patient's condition or delayed care. By providing timely and skilled management during transport, a pediatric transport team can enhance clinical outcomes, minimize potential complications, and improve overall patient safety.

Purpose:

The purpose of this policy is to establish clear, standardized criteria for the escorting medical team and use a standard checklist to ensure safe, efficient, and effective transfer of critically ill pediatric patients within the hospital.

Scope

The scope of this policy include all aspects of the transfer process for pediatric patients within the PICU and HD areas in the hospital. It applies to all healthcare personnel involved in the transport of critically ill pediatric patients, including:

- **Personnel:** The policy applies to all members of the transport team, including physicians, nurses, respiratory therapists and other medical staff involved in intra-hospital transport. It outlines the necessary criteria required for performance of the escort.
- **Transport Equipment and Resources:** The policy defines the required medical equipment and monitoring devices necessary for the safe transport of pediatric patients. This includes life-support equipment, medications, and other specialized tools specific to pediatric care.
- **Stabilization of Patient Condition:** The policy addresses the need to stabilize patient's condition prior to transport.

Definitions

- Intra-Hospital Transport: refers to the process of safely transferring critically ill pediatric patients within the hospital setting, from one area or department to another, such as from the PICU to the operating room, imaging department, or another specialized care unit.
- Escorting Medical Team: a group of healthcare professionals who accompany and care for critically ill pediatric patients during their transfer within the hospital.
- Transport Checklist: is a standardized list of essential tasks, equipment, and procedures to be reviewed and completed before and during the transport of a critically ill pediatric patient within the hospital.

Criteria for the Escorting Medical Team:

Physicians:

1. Should Hold pediatric advanced life support certification
2. Had worked in PICU or PHD for minimum of 3 months

Nursing Team:

1. Should Have a valid BLS or PALS certificates
2. Had worked in PICU or PHD for minimum of 3 months

Preparation for Intra-Hospital Transport:

Preparation of the Transport Equipment (Review Equipment Checklist)

Proper preparation of equipment is a critical component of ensuring the safe and effective transport of pediatric patients within the hospital. It is essential that all necessary equipment is thoroughly checked and readily available before initiating the transport. This process involves several key steps to ensure that the transport team is fully prepared to provide optimal care throughout the transfer.

1. Cardiac monitors: should be checked to ensure they are functioning properly.
2. Pulse oximeters and blood pressure cuffs should be calibrated and ready.
3. Endotracheal tubes, laryngoscopes, suction equipment, and other airway management tools should be available in various sizes suitable for the child's age and size.
4. Ventilators and oxygen delivery systems should be checked. Additional oxygen cylinders may be needed depending on the duration of the transport.

Preparation of the Escorting Team

Before the transport begins, the team should undergo a pre-transport briefing to review the patient's medical status, anticipated needs, and any potential risks during the transfer. This ensures that everyone involved understands the child's condition and the specific interventions that may be required. The team must also be familiar with the transport route and the facilities or departments involved, allowing for smooth transitions and clear communication during the journey.

Preparation of the Patient

Before transport begins, a thorough assessment of the patient's medical condition must be performed to identify any immediate issues or interventions required. This includes reviewing vital signs, respiratory status, and cardiovascular stability. The patient should be adequately stabilized, ensuring that any critical issues are addressed. Critical issues include stabilization of the ventilatory status ensuring an appropriate blood gas exchange. A blood gas should be checked prior to transport especially if the patient was changed to another ventilator or an adjustment of the ventilation parameters were made. In addition, the hemodynamic status should be optimized with fluids and inotropic support.

Another common issue is the administration of adequate sedation, analgesia and paralysis based on the child's condition.

TRANSPORT CHECKLIST FOR PEDIATRIC INTRA-HOSPITAL TRANSFER

(Note: This section should be filled by the physician)

1. Patient Details

Name: _____ | Age: _____ | MRN: _____
Weight: _____ | Diagnosis: _____
Reason/Indication for MRI/CT: _____ Date of Imaging _____
Urgency of Imaging: ☐ Routine ☐ Emergency

2. Organ Support

Respiratory Support

Is the patient ventilated? ☐ Yes ☐ No

If Yes, specify below:

Mode of Ventilation: ☐ Invasive ☐ Non-Invasive ☐ HFNC

FIO2: _____% PEEP: _____ | PIP: _____ | Rate: _____

Circulatory Support

Is the patient on inotropes? ☐ Yes ☐ No

If Yes, specify:

☐ Adrenaline ☐ Noradrenaline ☐ Milrinone ☐ Dopamine ☐ Others: _____

Vascular Access:

☐ Peripheral Lines, Site: _____ ☐ CVC, Site: _____

Sedation and Paralysis

Is the patient on sedation? ☐ Yes ☐ No

If Yes, specify:

☐ Fentanyl ☐ Morphine ☐ Dexmedetomidine ☐ Midazolam

Is the patient on paralysis? ☐ Yes ☐ No

3. Emergency Plans for Possible Complications during the Procedure

A. List sedatives/paralytic agents and doses if extra sedation is needed:

B. Inotropic plan if required:

C. Other anticipated emergencies:

4. Important Contact Numbers

- ☐ PICU Extension: _____
☐ PICU Registrar Pager: _____
☐ PICU Consultant on Service, Pager: _____
☐ Anesthesia Covering MRI Pager: _____

Note: This section should be filled by the Nursing)

5. Monitoring During MRI/Procedure

Monitoring every 10 minutes for ventilated patients:

[illegible]

	<u>Medical Equipment and Consumables</u>	Status		
		Qty	Yes	No
	1. Portable medication box and intravenous fluids	1	☐	☐
	2. Stethoscope (neonatal, pediatric)	1	☐	☐
Oxygen Accessories	2. Portable oxygen equipment			
	a) Portable oxygen cylinders minimum 300 liters capacity Note: Oxygen Cylinders type 48 are present in ambulance *Calculate double oxygen requirement for the journey as follows <i>Oxygen Requirement Calculation= 2x (Transport duration in minutes) x (O2 consumption in L/min)</i> <i>Oxygen Requirement Calculation for ventilated patients= 2X transport time in minutes x (minute ventilation XFio2)+ventilator driving gas</i>	1-4	☐	☐
	b) Constant flow regulator with adjustable flow rate (2-15 lpm)	1	☐	☐
	3. Oxygen administration equipment as applicable			
	a) Nasal cannula (pediatric)	1	☐	☐
	b) Non rebreathing mask (pediatric)	1	☐	☐
	c) Face mask	1	☐	☐
	d) Oxygen tubing (adequate length)	1	☐	☐
	4. Self-inflating Bag valve mask with oxygen reservoir (neonatal, pediatric)	1	☐	☐
Cardiac monitoring	5. Cardiac Monitoring			
	a) Cardiac monitor with full accessories (neonatal ,pediatric)	1	☐	☐
	b) ECG electrodes	1	☐	☐
Airway Equipment	6. Airways			
	a) Oro-pharyngeal airway sizes 0 – 3 <i>*Correct size measures from angle of mouth to angle of jaw</i>	1	☐	☐
	b) Laryngeal mask airway size 1 – 3 <i>*LMA size based on weight</i> Size 1 (<5 kg) Size 1.5 (5-10 kg) Size 2 (10-20 kg) Size 2.5(20-30 kg) Size 3 (30-50 Kg)	1	☐	☐
	7. Intubation Kit			
	a) Laryngoscope handles choose as applicable	1	☐	☐
	b) Laryngoscope blades as applicable (curved and straight) Neonatal /Pediatric Blades: Straight Blades 0,1,2,3. Curved Blades: 1,2,3,4	1	☐	☐
	c) Endotracheal tubes cuffed and/or un-cuffed Formula to calculate ET Tube size: <i>Uncuffed tubes= Age/4 +4</i> <i>Cuffed Tubes = Age/4 +3.5</i> <i>+ add one size smaller</i>	2	☐	☐
	d) Stylet for ET tubes and Magill forceps	1	☐	☐
	8. Ventilation and Airway			
	a) Portable ventilator, Cylinder, Oxygen regulator, test lung as applicable MRI Compatible as applicable Y/N Cylinder MRI compatible as applicable Y/N	1 1	☐ ☐	☐ ☐
Medications	9. Medications			
	Adrenaline 1:10,000 and 1:1000		☐	☐
	Atropine sulphate		☐	☐
	Sedation : Ketamine, Fentanyl, and paralytic agents,others		☐	☐
	Inotropes as applicable		☐	☐
	Fluid boluses as applicable		☐	☐

	Medical Equipment and Consumables	Status		
		Qty	Yes	No
	10. Portable medication box and intravenous fluids	1	☐	☐
	2. Stethoscope (neonatal, pediatric)	1	☐	☐
Oxygen Accessories	11. Portable oxygen equipment			
	b) Portable oxygen cylinders minimum 300 liters capacity Note: Oxygen Cylinders type 48 are present in ambulance *Calculate double oxygen requirement for the journey as follows <i>Oxygen Requirement Calculation= 2x (Transport duration in minutes) x (O2 consumption in L/min)</i> <i>Oxygen Requirement Calculation for ventilated patients= 2X transport time in minutes x (minute ventilation X Fio2)+ventilator driving gas</i>	1-4	☐	☐
	c) Constant flow regulator with adjustable flow rate (2-15 lpm)	1	☐	☐
	12. Oxygen administration equipment as applicable			
	e) Nasal cannula (pediatric)	1	☐	☐
	f) Non rebreathing mask (pediatric)	1	☐	☐
	g) Face mask	1	☐	☐
	h) Oxygen tubing (adequate length)	1	☐	☐
	13. Self-inflating Bag valve mask with oxygen reservoir (neonatal, pediatric)	1	☐	☐
	14. Cardiac Monitoring			
Cardiac monitoring	c) Cardiac monitor with full accessories (neonatal ,pediatric)	1	☐	☐
	d) ECG electrodes	1	☐	☐
Airway Equipment	15. Airways			
	c) Oro-pharyngeal airway sizes 0 – 3 <i>*Correct size measures from angle of mouth to angle of jaw</i>	1	☐	☐
	d) Laryngeal mask airway size 1 – 3 <i>*LMA size based on weight</i> Size 1 (<5 kg) Size 1.5 (5-10 kg) Size 2 (10-20 kg) Size 2.5(20-30 kg) Size 3 (30-50 Kg)	1	☐	☐
	16. Intubation Kit			
	e) Laryngoscope handles choose as applicable	1	☐	☐
	f) Laryngoscope blades as applicable (curved and straight) Neonatal /Pediatric Blades: Straight Blades 0,1,2,3. Curved Blades: 1,2,3,4	1	☐	☐
	g) Endotracheal tubes cuffed and/or un-cuffed Formula to calculate ET Tube size: <i>Uncuffed tubes= Age/4 +4</i> <i>Cuffed Tubes = Age/4 +3.5</i> <i>+ add one size smaller</i>	2	☐	☐
	h) Stylet for ET tubes and Magill forceps	1	☐	☐
	17. Ventilation and Airway			
	b) Portable ventilator, Cylinder, Oxygen regulator, test lung as applicable MRI Compatible as applicable Y/N Cylinder MRI compatible as applicable Y/N	1 1	☐ ☐	☐ ☐
Medications	18. Medications			
	Adrenaline 1:10,000 and 1:1000		☐	☐
	Atropine sulphate		☐	☐
	Sedation : Ketamine, Fentanyl, and paralytic agents,others		☐	☐
	Inotropes as applicable		☐	☐
	Fluid boluses as applicable		☐	☐

References:

Bernard Foex (2019), Neonatal Adult and Paediatric Safe Transfer and Retrieval
A Practical Approach to Transfers. John Wiley & Sons Ltd.